

LAB 3 – Component Modelling & Architectural Pattern Selection

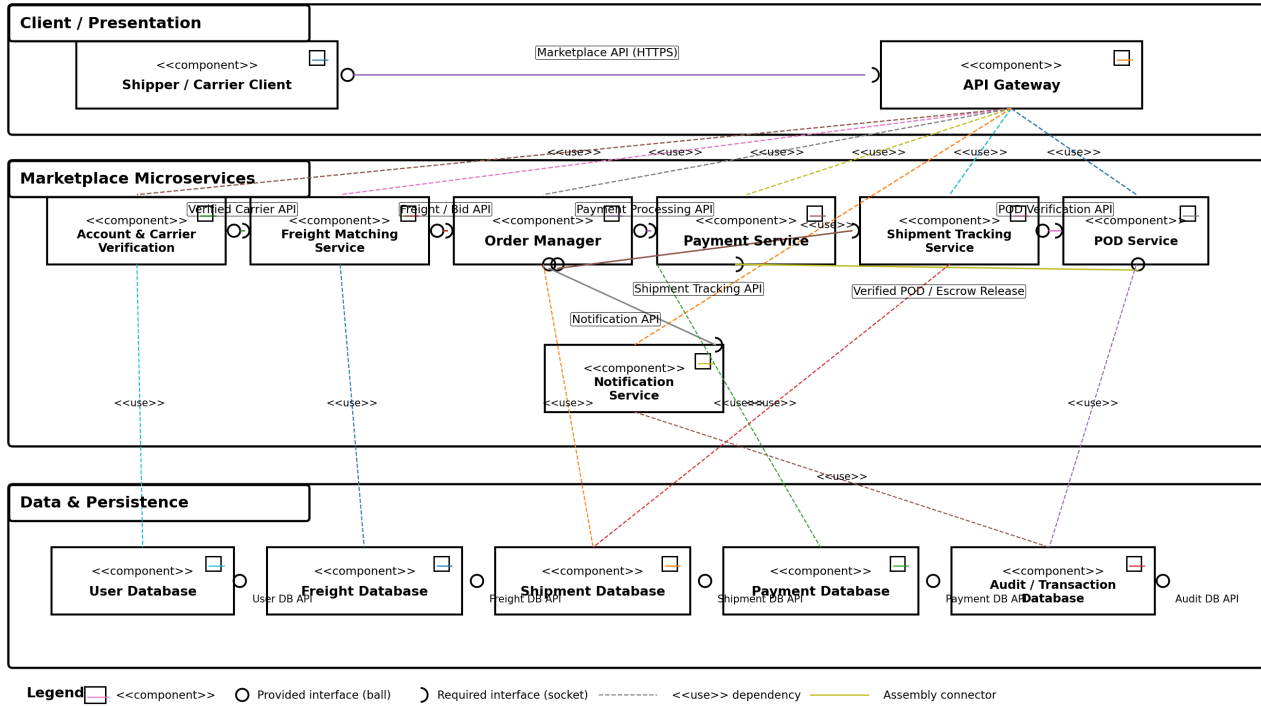
Project: Inter-City Freight Load Matching Marketplace (IFLM)

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DELIVERABLE 1 — UML COMPONENT DIAGRAM

Inter-City Freight Load Matching Marketplace (IFLM)

Component Diagram — Microservices Architecture



Selected Architecture: Microservices Architecture

Project functions represented: account/carrier verification, freight posting and competitive bidding, winning-carrier selection and shipment transaction management, shipment tracking, proof of delivery, escrow payment release, notifications, and transaction/audit data.

DELIVERABLE 2 — ARCHITECTURE SELECTION & JUSTIFICATION

Architecture Selection: Microservices Architecture

We chose Microservices Architecture for the Inter-City Freight Load Matching Marketplace. The system contains several logically independent functions, including account and carrier verification, freight posting and competitive bidding, shipment transaction management, shipment tracking, proof-of-delivery verification, escrow payment settlement, and transaction-data preservation.

Reason 1 – Independent scalability: Different parts of the marketplace can experience different workloads. For example, during an active freight auction, the Freight Matching Service may receive many requests for browsing loads and submitting bids. That service can be scaled independently without unnecessarily scaling payment or proof-of-delivery functions.

Reason 2 – Fault isolation and reliability: Bidding, shipment tracking, proof of delivery, and payment are separated into services. A failure in one service can therefore be isolated rather than automatically stopping the entire marketplace. This supports the requirement to preserve accepted bids, shipment records, milestone information, and payment transactions without data loss.

Security advantage: Sensitive payment operations can be isolated inside the Payment Service and accessed through controlled service interfaces. Authentication and authorization can restrict which components and users are allowed to invoke payment operations, reducing unnecessary exposure of payment data.

Performance benefit: Services can be independently scaled according to demand. High bidding activity can be handled by scaling the Freight Matching Service while unrelated services remain at their normal capacity. This helps the marketplace respond efficiently during peak activity.

Conclusion: Microservices Architecture is the most suitable choice for IFLM because it matches the system's independent business functions and provides independent scalability, fault isolation, security separation, and performance benefits.